

Time Series Analysis

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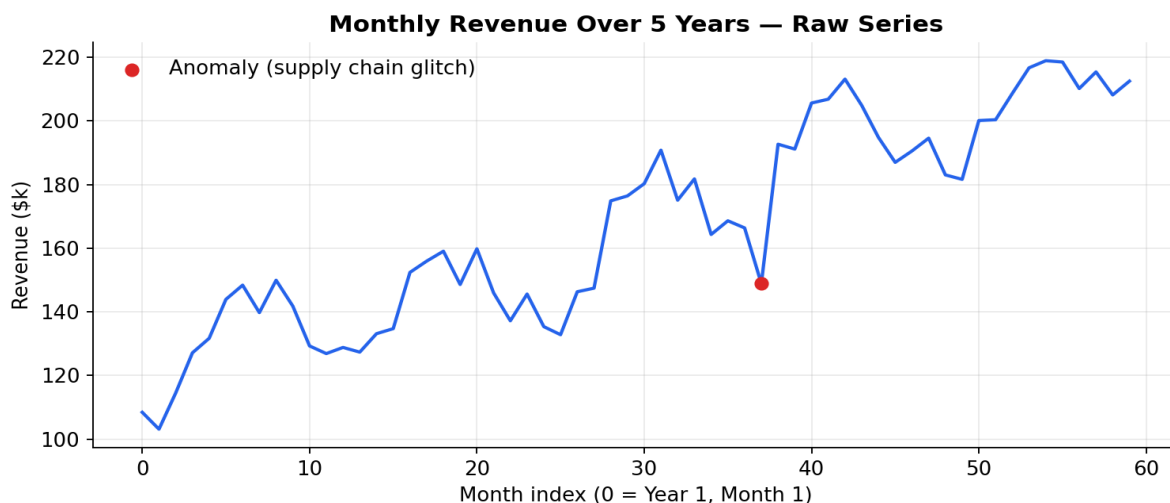
1. What Is a Time Series?

A **time series** is a sequence of data points recorded at successive, evenly spaced points in time. The feature that separates it from ordinary data is that **order matters** — each observation is tied to a specific point in time, and the sequence itself carries information about trends, cycles, and how past values influence future ones.

Examples of time series data:

- Daily closing stock prices
- Monthly sales revenue for a company
- Hourly temperature readings
- Annual GDP of a country
- Daily number of hospital admissions

Below is an example: five years of simulated monthly revenue for a retail company. Notice it isn't a smooth line — there's an overall upward drift, a repeating wiggle, and one sharp one-off drop.



The red dot marks a single anomalous month — a supply chain disruption that caused an unexpected dip. Spotting whether a movement like this is “real” or just noise is one of the central jobs of time series analysis.

2. What Is Time Series Analysis?

Time series analysis is the process of studying this kind of ordered data to understand its underlying structure, extract meaningful statistics, and build models that describe the past and forecast the future. It typically tries to answer:

- Is there a long-term trend (growing or shrinking)?
- Are there repeating seasonal patterns?

- How much of today's value depends on yesterday's (or last month's)?
- Can we predict next month's value with reasonable confidence?
- Was a sudden change a real shift or just random noise?

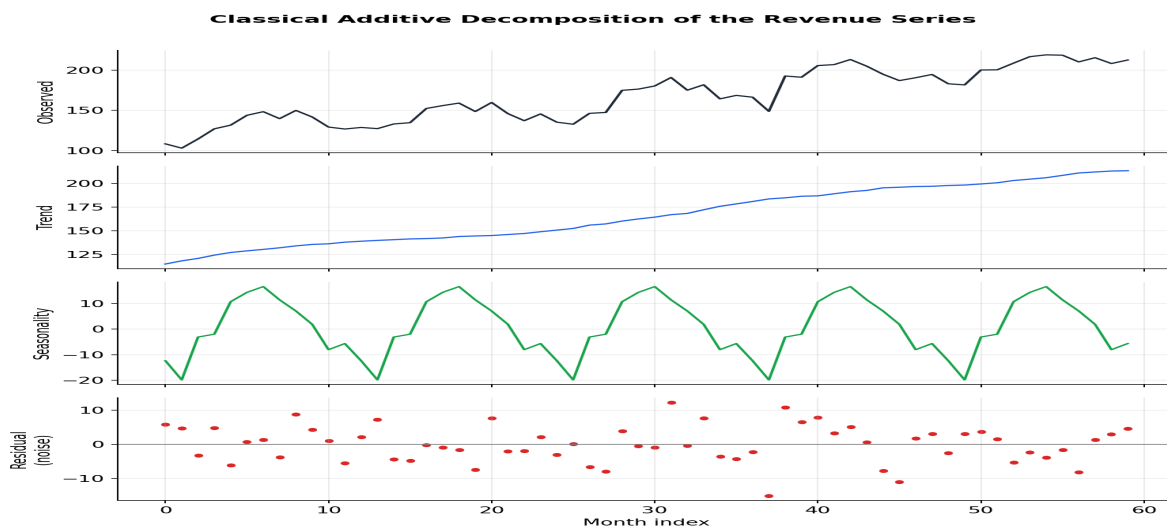
3. Core Components of a Time Series

Most time series can be broken down into four components:

Component	Description	Example
Trend	Long-term increase or decrease	Steadily rising smartphone sales over a decade
Seasonality	Regular pattern tied to a fixed calendar period	Ice cream sales spiking every summer
Cyclical	Fluctuation with no fixed period, tied to economic cycles	Boom-and-bust housing market cycles
Noise	Random variation left after removing the above	A one-day traffic dip from a server outage

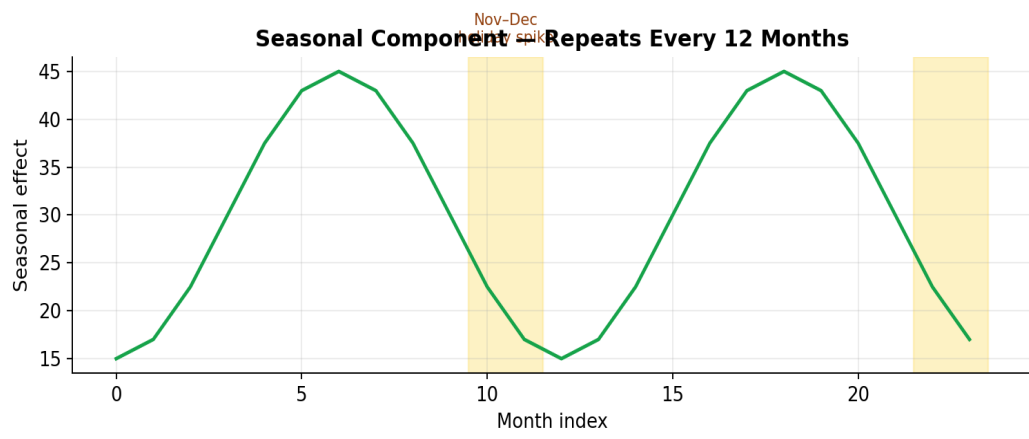
A common way to combine them is: **Observed = Trend + Seasonality + Cyclical + Noise** (an *additive* model), or their product in a *multiplicative* model, depending on whether seasonal swings grow along with the trend.

Decomposing the revenue example into these pieces makes the structure much clearer:



Once trend and seasonality are stripped away, the residual (bottom panel) should look like unpredictable noise scattered around zero — apart from the clear outlier around month 37, which lines up with the supply chain glitch.

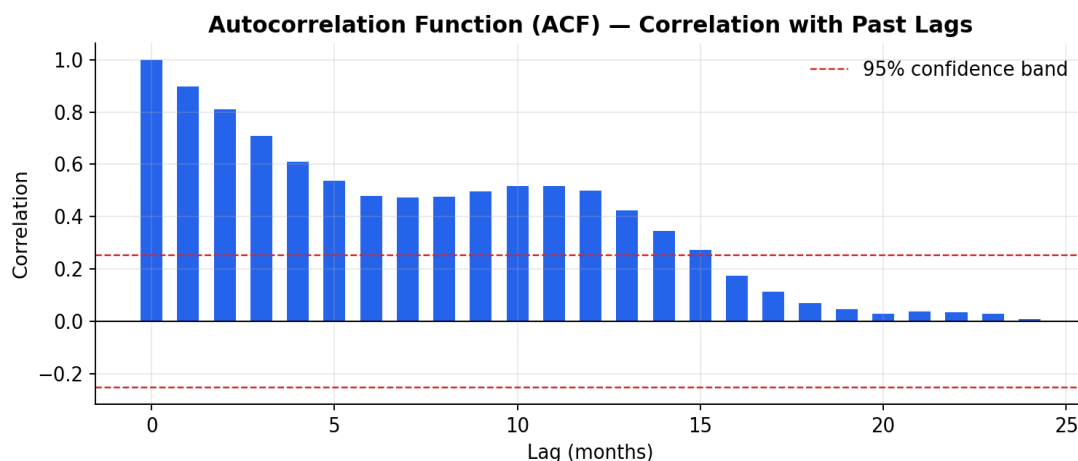
Zooming in on just the seasonal piece shows why it's called “seasonal” — it repeats identically every 12 months:



4. Key Concepts

- **Stationarity** — A series is *stationary* if its statistical properties (mean, variance) stay constant over time. Many classical models (like ARIMA) require this, so non-stationary data is often “differenced” to remove trend.
- **Autocorrelation** — The correlation of a series with a lagged (delayed) version of itself, e.g. does this month's revenue correlate with revenue 12 months ago?
- **Lag** — A previous time point used as a predictor. A “lag-12” feature for forecasting January uses January's value from the previous year.
- **White noise** — A series with no discernible pattern. If a model's residuals resemble white noise, it has captured all the predictable structure.

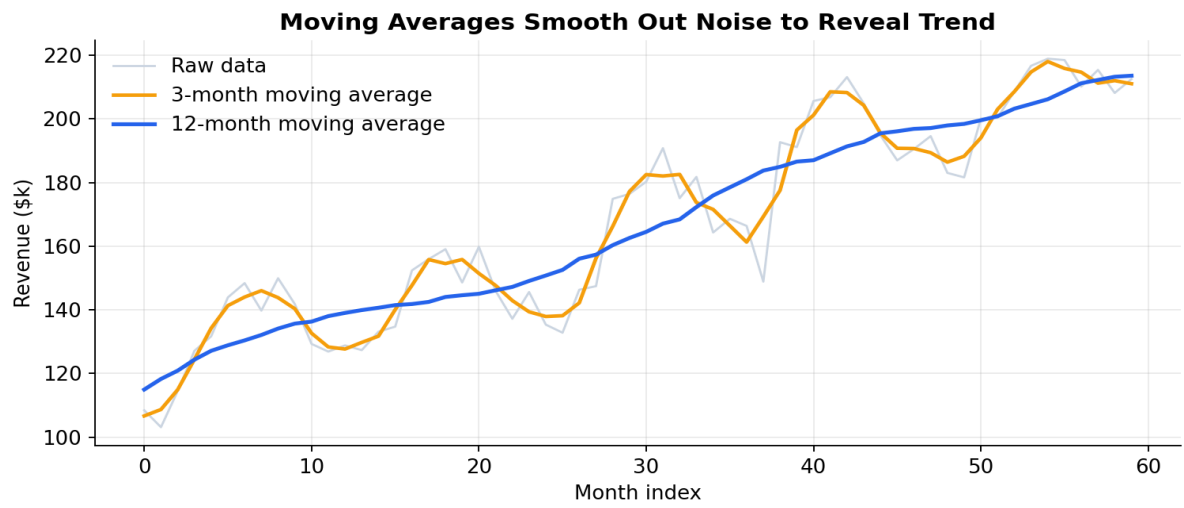
The Autocorrelation Function (ACF) plot below measures correlation at each lag for the revenue example. The tall spike near lag 12 (and its echo at 24) is the fingerprint of the yearly seasonal pattern:



5. Common Techniques and Models

- **Moving Averages** — smooth short-term fluctuations to reveal the underlying trend.
- **Exponential Smoothing (ETS)** — weights recent observations more heavily than older ones.
- **ARIMA** — combines autoregression, differencing, and moving averages of past errors; a classical forecasting workhorse.
- **SARIMA** — ARIMA extended with an explicit seasonal component.
- **Prophet** — built for business time series with strong seasonality, holidays, and missing data.
- **LSTM / Deep Learning** — neural networks for complex, non-linear or multivariate series.

The chart below shows a 3-month vs. 12-month moving average on the same data — the shorter window tracks local wiggles, the longer window isolates the trend:

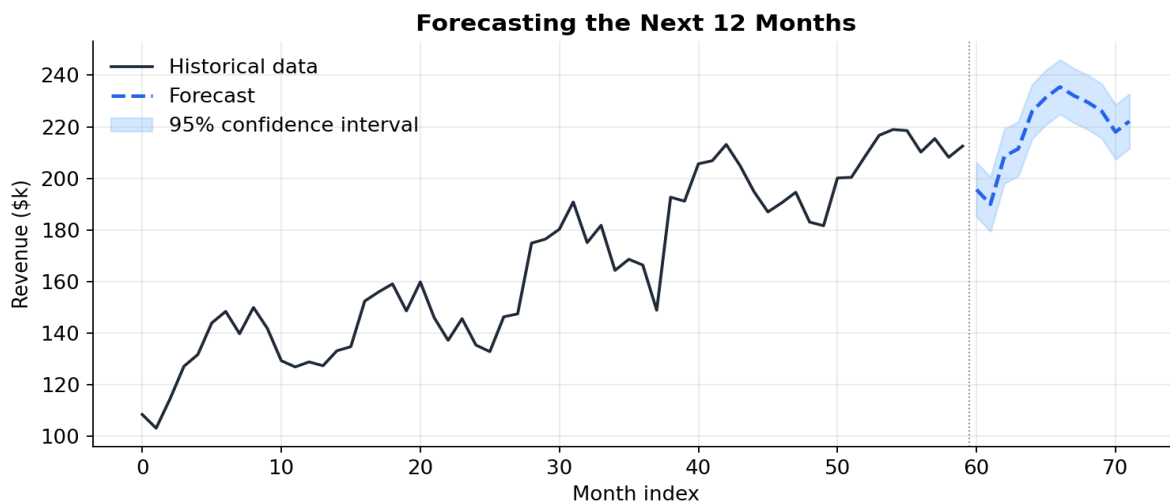


6. Real-World Example: Retail Revenue Forecast

Bringing it together — a retail company tracking monthly revenue over 5 years would typically observe:

- **Trend:** Revenue is generally increasing as the company grows.
- **Seasonality:** Revenue spikes every November–December due to holiday shopping.
- **Cyclical:** A broader multi-month dip tied to a regional economic slowdown.
- **Noise:** A random one-month drop caused by a supply chain glitch.

After decomposing the series and estimating trend and seasonality, both can be extended forward to produce a forecast, along with a confidence interval reflecting historical noise:



This lets the company separate the genuine holiday bump from underlying growth, plan inventory around the forecast, and tell whether a future dip is a real problem or just noise.

7. Common Applications

Domain	Typical Use
Finance	Stock price forecasting, volatility and risk modeling
Economics	GDP, inflation, and unemployment forecasting
Retail	Demand forecasting, inventory and staffing planning
Healthcare	Disease outbreak tracking, patient vital-sign monitoring
Weather	Temperature and precipitation forecasting
IoT / Operations	Server load monitoring, predictive maintenance

8. Summary

Time series analysis is fundamentally about respecting **order and time dependence** in data. By decomposing a series into trend, seasonality, cycle, and noise, and by measuring how values relate to their own past (autocorrelation), analysts can build models — from simple moving averages to ARIMA and deep learning — that explain what happened and forecast what's likely to happen next.

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